

Problem 1

Problem 1

(a) Derive the rotation matrix C for a rigid body in terms of the 3-1-3 Euler angles. These rotate the object first around the $\hat{n}_3$ -axis by an angle ψ , then around the new $\hat{b}_1$ -axis by an angle θ , and finally around the new $\hat{b}_3$ axis by an angle ϕ .

(b) Next, derive the matrix $B(\vec{\theta})$ that defines the *Kinematic Differential Equation* $\dot{\vec{\theta}} = B(\vec{\theta})\vec{\omega}$, where $\vec{\theta}$ is the vector $[\psi, \theta, \phi]$.

(a) Rotation matrix C

$$R(\psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}, \quad R(\phi) = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C = R(\phi) R(\theta) R(\psi)$$

Intermediate product:

$$R(\theta)R(\psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi \cos \theta & \cos \psi \cos \theta & \sin \theta \\ \sin \psi \sin \theta & -\cos \psi \sin \theta & \cos \theta \end{bmatrix}$$

$$C = \begin{bmatrix} \cos \psi \cos \phi - \sin \psi \cos \theta \sin \phi & \sin \psi \cos \phi + \cos \psi \cos \theta \sin \phi & \sin \theta \sin \phi \\ -\cos \psi \sin \phi - \sin \psi \cos \theta \cos \phi & -\sin \psi \sin \phi + \cos \psi \cos \theta \cos \phi & \sin \theta \cos \phi \\ \sin \psi \sin \theta & -\cos \psi \sin \theta & \cos \theta \end{bmatrix}$$

(b) Kinematic matrix B

Body-frame angular velocity is the sum of each Euler rate along the axis it turns about, each expressed in the body frame:

$$\begin{aligned} \vec{\omega}_B &= \dot{\phi} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \dot{\theta} R(\phi) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \dot{\psi} R(\phi)R(\theta) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ R(\phi) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} &= \begin{bmatrix} \cos \phi \\ -\sin \phi \\ 0 \end{bmatrix}, \quad R(\phi)R(\theta) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \sin \phi \sin \theta \\ \cos \phi \sin \theta \\ \cos \theta \end{bmatrix} \\ \vec{\omega}_B &= \underbrace{\begin{bmatrix} \sin \phi \sin \theta & \cos \phi & 0 \\ \cos \phi \sin \theta & -\sin \phi & 0 \\ \cos \theta & 0 & 1 \end{bmatrix}}_{B^{-1}} \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} \end{aligned}$$

Inverting B^{-1} to solve for $\dot{\vec{\theta}}$ in terms of $\vec{\omega}$:

$$B = \begin{bmatrix} \frac{\sin \phi}{\sin \theta} & \frac{\cos \phi}{\sin \theta} & 0 \\ \cos \phi & -\sin \phi & 0 \\ -\frac{\cos \theta \sin \phi}{\sin \theta} & -\frac{\cos \theta \cos \phi}{\sin \theta} & 1 \end{bmatrix}$$

Problem 2

Problem 2

Please model the dynamics of an axisymmetric spinning top, spinning about a point fixed P on the ground. To do so, we will use the 3-1-3 Euler angles that you derived in Problem 1. Because the heavy top is axisymmetric, we can take the moments of inertia with respect to P to be $I_t = I_1 = I_2$ and $I_a = I_3$, and choose the principal axes centered at P as our body-fixed axes $\{\hat{b}\}$. The center of mass is a distance l away from P along the b_3 -axis

(a) Write the kinetic and potential energies of the top to find the Lagrangian, and then derive Lagrange's equations of motion.

(b) Two of the three Euler angles are ignorable. Which ones?

(c) Use the Routhian procedure that we've used in the past to get an equation of motion for the only non-ignorable coordinate. Define the constants $a = B_1/I_1$ and $b = B_2/I_2$, with $B_i = \partial L/\partial q_i$. Find the 2nd order equation of motion for the non-ignorable coordinate, and the first-order equations of motion for the other two.

(a) Lagrangian and equations of motion

From (b), with $\omega_3 = \cos \theta \dot{\psi} + \dot{\phi}$:

$$\omega_1 = \sin \phi \sin \theta \dot{\psi} + \cos \phi \dot{\theta}, \quad \omega_2 = \cos \phi \sin \theta \dot{\psi} - \sin \phi \dot{\theta}$$

Cross terms cancel in $\omega_1^2 + \omega_2^2 = \sin^2 \theta \dot{\psi}^2 + \dot{\theta}^2$, so with $T = \frac{1}{2} \vec{\omega}^T I_P \vec{\omega}$:

$$T = \frac{1}{2} I_t (\sin^2 \theta \dot{\psi}^2 + \dot{\theta}^2) + \frac{1}{2} I_a \omega_3^2, \quad V = MgL \cos \theta$$

$$\mathcal{L} = T - V = \frac{1}{2} I_t (\sin^2 \theta \dot{\psi}^2 + \dot{\theta}^2) + \frac{1}{2} I_a (\cos \theta \dot{\psi} + \dot{\phi})^2 - MgL \cos \theta$$

$$\phi: \frac{\partial \mathcal{L}}{\partial \phi} = 0 \text{ (ignorable); } \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = I_a \omega_3 = \text{const}$$

$$\psi: \frac{\partial \mathcal{L}}{\partial \psi} = 0 \text{ (ignorable); } \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = I_t \sin^2 \theta \dot{\psi} + I_a \omega_3 \cos \theta = \text{const}$$

$$\theta: \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = I_t \ddot{\theta}, \quad \frac{\partial \mathcal{L}}{\partial \theta} = I_t \sin \theta \cos \theta \dot{\psi}^2 - I_a \omega_3 \sin \theta \dot{\psi} + MgL \sin \theta$$

$$\ddot{\theta} = \sin \theta \cos \theta \dot{\psi}^2 - \frac{I_a}{I_t} \omega_3 \sin \theta \dot{\psi} + \frac{MgL}{I_t} \sin \theta$$

(b) Ignorable coordinates

ψ and ϕ are ignorable: $\mathcal{L}$ has no explicit dependence on either.

(c) Routhian reduction

Let $B_\psi = \partial \mathcal{L} / \partial \dot{\psi}$, $B_\phi = \partial \mathcal{L} / \partial \dot{\phi}$ (both constant, from (a)), and define

$$a = \frac{B_\psi}{I_t} = \sin^2 \theta \dot{\psi} + b \cos \theta, \quad b = \frac{B_\phi}{I_t} = \frac{I_a}{I_t} \omega_3$$

$$\dot{\psi} = \frac{a - b \cos \theta}{\sin^2 \theta}$$

From $\omega_3 = \cos \theta \dot{\psi} + \dot{\phi} = (I_t/I_a)b$:

$$\dot{\phi} = \frac{I_t b}{I_a} - \frac{a \cos \theta - b \cos^2 \theta}{\sin^2 \theta}$$

Substituting $\dot{\psi}$ into the θ equation of (a) (with b in place of $\frac{I_a}{I_t}\omega_3$):

$$\ddot{\theta} = \frac{\cos \theta (a - b \cos \theta)^2}{\sin^3 \theta} - \frac{b(a - b \cos \theta)}{\sin \theta} + \frac{MgL}{I_t} \sin \theta$$